IVY X. HE

Email: xiao_he@brown.edu | **Cell:** 832-679-6589 | **Website:** ivyyyy24381.github.io/

Research Focus: *Multimodal human-robot interaction, Embodied AI, probabilistic decision-making (POMDP), cognitive robotics, and interdisciplinary approaches combining art, robotics, and cognitive science.*

Brown University	Providence, RI	2023-Present
3rd year Ph.D. Candidate in Computer Science	*Advisor: Professor S. Tellex*	*GPA: 4.00/4.00*
Carnegie Mellon University	Pittsburgh, PA	2019-2023
B.Sc. in Mechanical Engineering	*Additional Major in Robotics*	*GPA: 3.91/4.00*

ONGOING RESEARCH EXPERIENCE

FetchHound: learning social attention from human–canine to human–robot object search

Embodied AI & Trustworthiness in HRI — *Advised by Prof. S. Tellex* — 2026–Now

- Developed a novel pipeline to teach a quadruped's attention-driven, animal-like behavior via LfD, generating training data by reconstructing 3D gaze and attention from multi-camera dog video.
- Built the end-to-end simulation system from 3D reconstruction to motion tracking and closed-loop policy roll-out (PyTorch MLP/LSTM/Transformer) in Unity.

Preliminary results presented at ICRA2026 Workshop

UMBRA: A Multi-Robot Shadow Performance System

Multi-Robot Coordination & Motion Planning — *Advised by Profs. A. Schulz & S. Tellex* — 2026–Now

- Coordinated a fleet of SO-101 arms to cast target silhouettes; built system pipeline including analytic pose solving, prop assignment, base-placement search, & 5-DOF IK in MuJoCo and deployed to real hardware.
- Turned target videos into per-frame joint trajectories via CMA-ES and a JAX gradient solver with analytic shadow Jacobians; shipped a Next.js + FastAPI fleet-control app with CI-tested Python (pytest, ruff)

Project supported by the National Endowment for the Arts(NEA)

RELEVANT PUBLICATIONS

LEGS-POMDP: Language and Gesture Conditioned Object Search

Multimodal Instruction Grounding in HRI — *Advised by Prof. S. Tellex* — 2024–2025

- Designed a POMDP-based framework that fuses 3D gesture cones, language prompts, and visual grounding as a joint observation model for robust multimodal object search.
- Multimodal input (gesture + language) outperformed single modalities by over 23%, achieving 76.6% success; an IRB-exempted user study is in progress to assess interaction usability.

HRI 2026 Full Paper; ICRA 2025 Late Breaking Results

Unitac: Whole-Robot Touch Sensing Without Tactile Sensors

Tactile Sensing in HRI — *Advised by Profs. S. Sridhar & S. Tellex* — 2024–2025

- Designed and implemented the Human–Robot Interaction (HRI) component, including touch-driven interaction paradigms and behavior design.
- Developed and evaluated interaction scenarios leveraging whole-body touch localization for expressive robot response on quadruped and manipulation platforms.

Presented at RSS 2025 HRCM workshop

Find it Like a Dog: Using Gesture to Improve Object Search

HRI–Cognitive Science Research — *Advised by Profs. S. Tellex & D. Buchsbaum* — 2023–2024

- Conducted a comparative study on five pointing vector representations to model human intent, integrating gesture understanding into a POMDP-based robot object search framework.

- Demonstrated that a combined arm-based vector representation significantly improves search accuracy; system validated through real-robot experiments and cross-species evaluation with domestic dogs.

Published in Cognitive Science Society Conference (CogSci 2024); presented at RSS 2024 TaskSpec Workshop

Social Environment for Autonomous Navigation 2.0 (SEAN 2.0)

Social Navigation & Simulation Research — *Advisors: Profs. H. Admoni & A. Steinfeld* — 2021-2023

- Assisted with building and curating interactive social navigation scenarios in the SEAN simulator using ROS and Unity integration with real-world pedestrian datasets. Designed dynamic restaurant environments and legibility metrics to evaluate robot behavior and human-robot interaction quality.

OTHER PUBLICATIONS

- *Rajko, J., Iyengar, V., Merlin, M., He, I., Lennington, E. (2026).* Choreographic and Improvisational Approaches To Interrogating Robotic Systems. *ACM MOCO 2026.*
- *Wang, K., He, I., Li, J., Asadipour, A., Sun, Y. (2024).* Exploring fungal morphology simulation and dynamic light containment from a graphics generation perspective. *SIGGRAPH Asia Art Papers.*
- *Wang, X., Ghaffarizadeh, A., He, I., & McGaughey, A. J. H., Malen, J., (2022).* Ultrahigh evaporative heat transfer measured locally in submicron water films. *Scientific Reports, 12(1), 22353.*

LEADERSHIP

PotBot: Autonomous Pothole Detection and Repair Robot

Lead Systems Engineer & Documentation Lead — *CMU Robotics Major Capstone* — 2022–2023

- Led full system design and team coordination, including mechanical integration, perception pipeline, and project planning.
- Developed and fine-tuned a YOLOv5 model for pothole detection using a custom dataset.

Iris Lunar Rover - the First university-built American lunar nano mobile robot

Systems Integration & Creative Lead — *CMU Space robotics* — 2020–2023

- Systematized Do-No-Harm testing documentation and designed Verification and Validation Matrix, Corrective Action Plan & Deviation Report for testing. Performed Failure Mode and Effects Analysis and assisted in the design of the Mission Operation Guide.
- Led media and branding efforts, including merchandise design, website, and community outreach.

SERVICE & OUTREACH

• *Providence Waterfire Robot Activation Booth*	Brown Booth Organizer	Aug. 2026
• *RoboBoston: 8th Annual Robot Block Party*	Brown Booth Organizer	Sep. 2025
• *Conference for Research on Choreographic Interfaces 2025*	Workshop Organizer	Jun. 2025
• *Point it Out: Using Gesture to Improve Object Search*	Honors Thesis Mentor	2023-2024
• *IEEE ICRA 2026, RSS2026, IROS2026, IROS 2025, ACM2024*	Reviewer	2024 -2026
• *Bristol BookFest 2025*	Speaker	2025
• *Veritas AI Fellowship: High School Research Program*	Research Mentor	2023-Now
• *MindMatters: LLM-powered Teen Wellness Support Platform*	Technical Mentor	2025-Now
• *CMU Student Academic Success Center*	Academic Coach	2020-2021

INTERNSHIPS

Argo AI, Integration & Testing Engineer Intern — Pittsburgh, PA |Summer 2022

- Designed, conducted, and analyzed Lidar Probability of Detection Test to evaluate lidar performance overtime & testings for radar and camera sensors, calibration system, and cleaning system.

RE2 Robotics, Mechanical Engineering Intern — Pittsburgh, PA |Summer 2020

- Compiled the Initial release of the User Manuals, inspected and assembled prototype joints for testing.